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Roll No

IT-5004-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Digital Communication

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) “Pulse modulation system is not digital, whereas pulse code modulation is” Justify.
b) What is quantization error? How does it depends upon the step size? Explain with suitable examples.
2. a) Draw block diagram of PCM system and explain its working. Define intersymbol interference and eye-patterns.
b) 24 telephone channels, each band limited to 3.4 KHz are to be time division multiplexed by using PCM. Calculate the bandwidth of the PCM system for 128 quantization levels and an 8KHz sampling frequency.
3. a) Explain frequency shift keying. Describe coherent detection of FSK signals. What should be the relationship between bit rate and frequency shift for a better performance.
b) Explain PSK and DPSK. Compare the two.

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4. a) Draw block diagram of QPSK transmitter. How is it better than PSK system.
b) Write a short note on MODEM.
5. a) What is Entropy? Show that the entropy is maximum when all the symbols are equi-probable. Assume $M = 3$.
b) The voice frequency modulating signal of a PCM system is quantized in 16 levels with following probabilities.
 $P_1 = P_2 = P_3 = P_4 = 0.1$, $P_5 = P_6 = P_7 = P_8 = 0.05$
 $P_9 = P_{10} = P_{11} = P_{12} = 0.075$ $P_{13} = P_{14} = P_{15} = P_{16} = 0.025$
Calculate the entropy and information rate.
Assume $F_m = 3\text{KHz}$.
6. a) What is channel capacity? Write equation for channel capacity of symmetric channel.
b) Write and explain Shannon Hartley theorem for capacity of Gaussian Channel.
7. a) Apply Shannon Fano coding procedure for the following message ensemble.
 $[X] = [x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7]$
 $[P] = [0.4 \quad 0.2 \quad 0.12 \quad 0.08 \quad 0.08 \quad 0.08 \quad 0.04]$
Find coding efficiency. Assume $M = 2$.
b) Differentiate cyclic and linear block codes.
8. Write short notes on any two:
 - i) Simplex and Duplex transmission
 - ii) NRZ codes
 - iii) Bipolar Line codes
 - iv) Convolutional codes

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